HYPER-LOCAL AVIATION WEATHER FORECAST VERIFICATION SYSTEM FOR SRI LANKAN MONSOON PATTERNS

30°C

MAX. 31°C - MIN.15°C

20%

15 KM/H

WEST

1023.0 MB

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PROBLEM

- Global weather models provide regional-scale forecasts (Ex: GFS and ECMWF)
- Airport weather conditions can change rapidly within a small area
- Monsoon-related variations are difficult to capture at airport level
- Existing forecasts may not accurately represent hyper-local conditions at VCBI

STAKEHOLDERS

- Pilots – weather information for flight decisions
- Air Traffic Controllers (ATC) – operational and safety decisions
- Airport Meteorological Officers – monitoring and forecasting
- Airport Operations – operational planning
- Civil Aviation Authorities – aviation safety and weather standards

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TEMPERATURE -PRESSURE COUPLED PREDICTION MODEL

RESEARCH GAP

identified is that global weather models like GFS, ECMWF do not provide the hyper-local accuracy required for VCBI, and they lack automated adaptability to changing local patterns.

OBJECTIVE

- Forecast Temperature, Pressure at vcbi
- Predict weather conditions for the next 3 hours
- Model the coupled relationship between the weather parameters
- Calculate Density Altitude
- Improve hyper-local aviation weather assessment at Bandaranaike International Airport

SOLUTION OVERVIEW

- Predict Temperature and Pressure at VCBI.
- Generate short-term T+3 hour predictions.
- Use a Hybrid LSTM + Random Forest Architecture.
- Calculate Pressure Altitude and Density Altitude, and classify conditions as Normal, Elevated, or High
- Support forecasters with human-readable aviation weather information.

Wind Pattern Prediction Model (speed and direction)

RESEARCH GAP

identified is that global weather models like GFS, ECMWF do not provide the hyper-local accuracy required for VCBI, and they lack automated adaptability to changing local patterns.

OBJECTIVE

- Predict Wind Speed and Wind Direction at VCBI.
- Generate 1-hour and 3-hour short-term predictions.
- Capture non-linear relationships between meteorological parameters.
- Provide runway-oriented headwind and crosswind information.
- Support forecasters with human-readable aviation weather information.

SOLUTION OVERVIEW

- Predict Wind Speed and Wind Direction at VCBI.
- For aviation operations,wind prdiction model is usable if its prediction error is less than 2.5 knots.
- Generate 1-hour and 3-hour short-term predictions use XGBOOST and TFT (Temporal Fusion Transformer) coupled prediction
- Continues learning Automated MLOps retraining using newly collected weather data
- Capture non-linear relationships between meteorological parameters.
- Provide runway-oriented headwind and crosswind information.

CLOUD AND VISIBILITY PREDICTION MODEL

RESEARCH GAP

- dentified is that global weather models like GFS, ECMWF do not provide the hyper-local accuracy required for VCBI, and they lack automated adaptability to changing local patterns.

OBJECTIVE

- Predict Cloud Status and Visibility
- Generate short-term airport-level predictions
- Capture non-linear relationships between meteorological parameters
- Provide human-readable aviation weather information

SOLUTION OVERVIEW

- Evaluated Decision Tree, Random Forest, and XGBoost
- XGBoost selected as the final model
- Separate models for Cloud Status and Visibility
- Uses METAR observations and engineered meteorological features
- Real-time prediction through Flask API
- Automated MLOps retraining using newly collected weather data
- Provides predictions in a human-readable aviation format (Cloud - FEW, SCT,BKN)

QNH, DEWPOINT AND RH PREDICTION MODEL

RESEARCH GAP

Traditional numerical weather prediction models lack the hyper-local precision required for safe aviation operations, making them unreliable in the face of Sri Lanka's highly volatile and rapidly shifting monsoon patterns, particularly when forecasting critical parameters like Atmospheric Pressure (QNH) and Dewpoint.

OBJECTIVE

- Forecast QNH and Dewpoint accurately
- Predict weather conditions for the next 3 hours
- Calculate Relative Humidity from predicted weather parameters

SOLUTION OVERVIEW

- Use LightGBM for fast and accurate weather forecasting
- Derive Relative Humidity using the August-Roche-Magnus formula
- Apply advanced feature engineering for weather patterns
- Use monthly automated MLOps retraining with live SQLite data
- Verify prediction accuracy through real-time monitoring

SYSTEM INTEGRATION

- Historical VCBI weather data → Preprocessing & Feature Engineering
- Monsoon-specific ML/DL components → 3 Hour Forecasting
- Forecast outputs → METAR / TAF Verification
- Verification results → Error & Reliability Analysis
- Integrated outputs → Pilot Weather Guidance / Decision Support

COMPLETION STATUS

- Individual prediction components developed
- Model outputs integrated into the common forecasting framework
- Forecast verification workflow integrated
- Overall system testing, validation and performance evaluation ongoing

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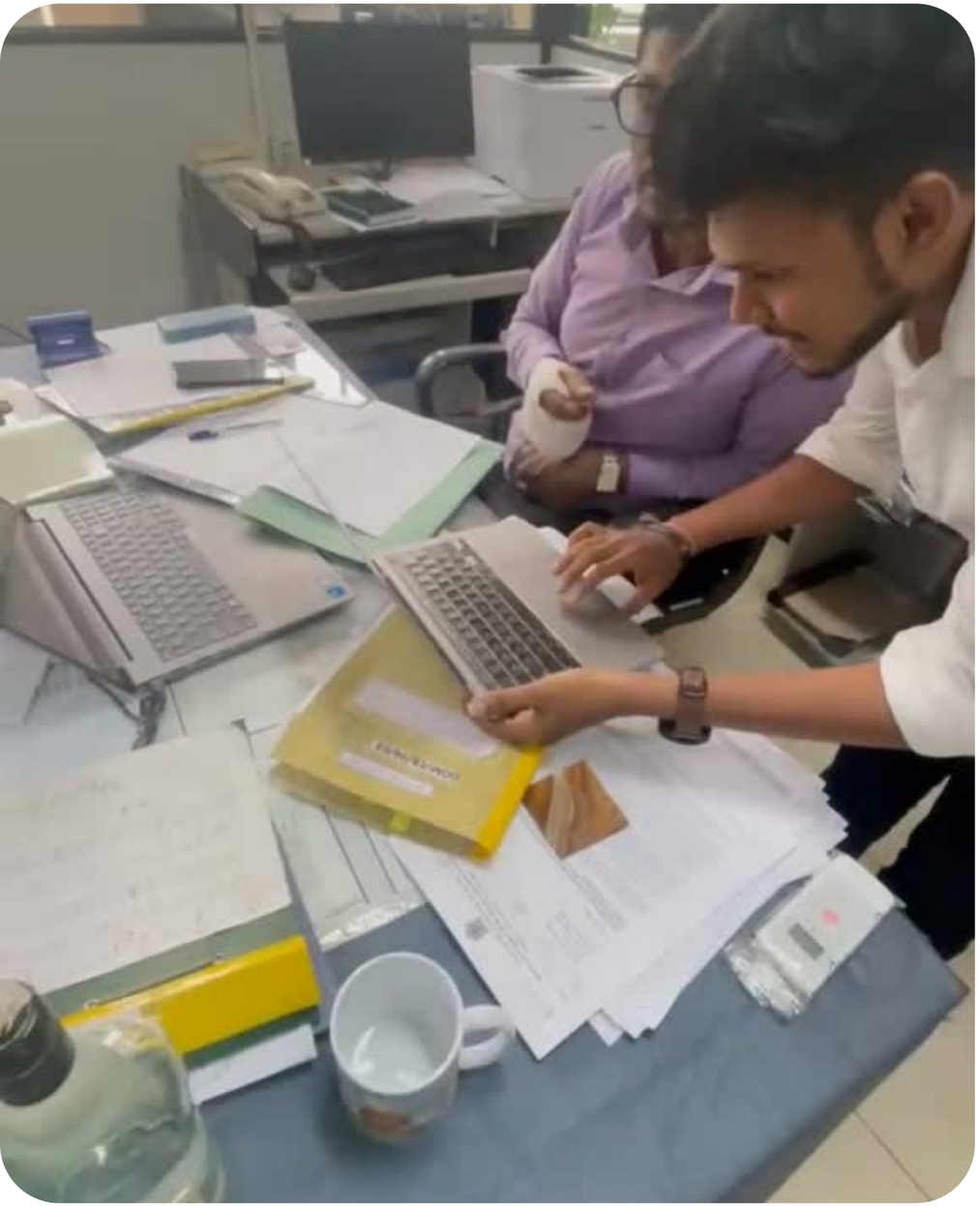

EVALUATION EVIDENCE

- Real VCBI weather data used
- Forecast compared with METAR/SPECI observations
- MAE, RMSE & R^2 used for evaluation
- Monsoon-specific model performance evaluated
- Forecast errors used for reliability assessment

NON-FUNCTIONAL REQUIREMENTS

- Accuracy – reliable T+3 forecasts
- Performance – fast forecast generation
- Reliability – consistent forecast & verification
- Usability – clear Pilot Weather Guidance
- Scalability – support future VCBI data
- Maintainability – continuous model retraining
- Security – protect stored forecast records

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COMMERCIALIZATION & DEPLOYMENT POTENTIAL

- **Internal Airport Use – Localized weather monitoring and forecasting for airport operations.**
- **Airline Operations – Support for flight planning and operational decision-making.**
- **Operational Decision Support – Help pilots, dispatchers, and airport staff make weather-related decisions.**
- **Future Deployment – Potential integration with the Department of Meteorology, subject to validation and approval.**

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Than k You